NutriPATH
INTEGRATIVE PATHOLOGY SERVICES
P: 1300 688 522
E: info@nutripath.com.au

BENJAMIN TRENGROVE
03-Oct-1989 Male

1 INTERNATIONAL PATIENT
MELBOURNE VIC 3000

Dr.RAJIV JOSHI
NUTRISEARCH PRACTITIONER
PO BOX 8177
HAWKES BAY NEW ZEALAND 1000

LAB ID : 3886986
UR NO. : 6190621
Collection Date : 24-Apr-2023
Received Date: 01-May-2023

3886986

COMPLETE MICROBIOME MAPPING

General Macroscopic Description

	Result	Markers
Stool Colour	Brown	**Colour** - Brown is the colour of normal stool. Other colours may indicate abnormal gut health.
Stool Form	Formed	**Form** -Sample form is categorised using the Bristol stool chart. A comment on stool appearance can be found in the comments section.
Mucous	Not Detected	**Mucous** - Mucous production may indicate the presence of an infection and/or inflammation.
Occult Blood	Negative	**Blood (Macro)**- The presence of blood in the stool may be the result of several causes besides colorectal bleeding, including hemorrhoids or gastrointestinal infection.

Short Chain Fatty Acids	Result	Range	Units
Short Chain Fatty Acids, Beneficial	48.3	> 13.6	umol/g
Butyrate	16.3	10.8 - 33.5	%
Acetate	64.1	44.5 - 72.4	%
Propionate	17.3	0.0 - 32.0	%
Valerate	2.2	0.5 - 7.0	%

GIT Functional Markers	Result	Range	Units
Calprotectin.	47.0	0.0 - 50.0	ug/g
Pancreatic Elastase	*147.0* *L	> 200.0	ug/g
Faecal Secretory IgA	*107.2* *L	510.0 - 2010.0	ug/g
Faecal Zonulin	*109.0* *H	0.0 - 107.0	ng/g
Faecal B-Glucuronidase	*4620.0* *H	337.0 - 4433.0	U/g
Steatocrit	13.0	0.0 - 15.0	%
anti-Gliadin IgA	<20	0.0 - 100.0	units/L

Microbiome Mapping Summary

Parasites & Worms

Bacteria & Viruses

Methanobacteriaceae
Enterotoxigenic E. coli LT/ST

Fungi and Yeasts

Key Phyla Microbiota

Firmicutes:Bacteroidetes Ratio	0.34	< 1.00	RATIO

Relative Commensal Abundance of the 6 Phyla groups can be found on page 4 of this report

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Parasites and Worms.	Result	Range	Units
Parasitic Organisms			
Cryptosporidium.	<dl	< 1.0	x10^6 org/g
Entamoeba histolytica.	<dl	< 1.0	x10^4 org/g
Giardia intestinalis	<dl	< 1.0	x10^3 org/g
Blastocystis hominis.	<dl	< 1.0	x10^3 org/g
Dientamoeba fragilis.	<dl	< 1.0	x10^5 org/g
Endolimax nana	<dl	< 1.0	x10^4 org/g
Entamoeba coli.	<dl	< 5.0	x10^6 org/g
Pentatrichomonas hominis	<dl	< 1.0	x10^2 org/g

Worms

Ascaris lumbricoides, Roundworm	Not Detected	Necator americanus, Hookworm	Not Detected
Trichuris trichiura, Whipworm	Not Detected	Enterobius vermicularis,Pinworm	Not Detected
Enterocytozoon spp	Not Detected	Hymenolepis spp, Tapeworm	Not Detected
Strongyloides spp, Roundworm	Not Detected	Taenia species, Tapeworm	Not Detected

Comment: Not Detected results indicate the absence of detectable DNA in the sample for the worms reported.
NOTE: Reflex testing is performed on clinically indicated samples

Opportunistic Bacteria/Overgrowth	Result	Range	Units
Bacillus species.	<dl	< 1.00	x10^5 CFU/g
Enterococcus faecalis	<dl	< 1.00	x10^4 CFU/g
Enterococcus faecium	<dl	< 1.00	x10^4 CFU/g
Morganella species	<dl	< 1.00	x10^3 CFU/g
Pseudomonas species	<dl	< 1.00	x10^4 CFU/g
Pseudomonas aeruginosa.	<dl	< 3.00	x10^2 CFU/g
Staphylococcus species	<dl	< 1.00	x10^4 CFU/g
Staphylococcus aureus	<dl	< 5.00	x10^2 CFU/g
Streptococcus species	2.30	< 3.00	x10^3 CFU/g
Methanobacteriaceae	*9.40* *H	< 5.00	x10^9 CFU/g
Desulfovibrio piger	<dl	< 18.00	x10^7 CFU/g
Oxalobacter formigenes	133.97	> 15.00	x10^7 CFU/g
Potential Autoimmune Triggers			
Citrobacter species.	<dl	< 5.00	x10^5 CFU/g
Citrobacter freundii.	<dl	< 5.00	x10^5 CFU/g
Klebsiella species	<dl	< 5.00	x10^3 CFU/g
Klebsiella pneumoniae.	<dl	< 5.00	x10^4 CFU/g
Prevotella copri	<dl	< 1.00	x10^7 CFU/g
Proteus species	<dl	< 5.00	x10^4 CFU/g
Proteus mirabilis.	<dl	< 1.00	x10^3 CFU/g
Fusobacterium species	0.80	< 10.00	x10^7 CFU/g

Fungi & Yeast	Result	Range	Units
Candida species.	<dl	< 5.00	x10^3 CFU/g
Candida albicans.	<dl	< 5.00	x10^2 CFU/g
Geotrichum species.	<dl	< 3.00	x10^2 CFU/g
Saccharomyces cerevisiae.	<dl	< 3.00	x10^3 CFU/g
Rhodotorula species.	<dl	< 1.00	x10^3 CFU/g

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Bacterial Pathogens	Result	Range	Units
Aeromonas hydrophila.	<dl	< 1.00	x10^3 CFU/g
Campylobacter species.	<dl	< 1.00	x10^3 CFU/g
C. difficile, Toxin A	<dl	< 1.00	x10^3 CFU/g
C. difficile, Toxin B	<dl	< 1.00	x10^3 CFU/g
Enterohemorrhagic E. coli	<dl	< 1.00	x10^3 CFU/g
Enteroinvasive E. coli/Shigella	<dl	< 1.00	x10^3 CFU/g
Enterotoxigenic E. coli LT/ST	*1.36* *H	< 1.00	x10^3 CFU/g
Shiga-like Toxin E. coli stx1	<dl	< 1.00	x10^3 CFU/g
Shiga-like Toxin E. coli stx2	<dl	< 1.00	x10^3 CFU/g
Salmonella species.	<dl	< 1.00	x10^4 CFU/g
Vibrio species.	<dl	< 1.00	x10^5 CFU/g
Yersinia species.	<dl	< 1.00	x10^5 CFU/g
Helicobacter pylori	<dl	< 1.0	x10^3 CFU/g

Comment: Helico Pylori virulence factors will be listed below if detected POSITIVE

H.pylori Virulence Factor, babA	Not Detected	H.pylori Virulence Factor, cagA	Not Detected
H.pylori Virulence Factor, dupA	Not Detected	H.pylori Virulence Factor, iceA	Not Detected
H.pylori Virulence Factor, oipA	Not Detected	H.pylori Virulence Factor, vacA	Not Detected
H.pylori Virulence Factor, virB	Not Detected	H.pylori Virulence Factor, virD	Not Detected

Viral Pathogens	Result	Range	Units
Adenovirus 40/41	Not Detected		
Norovirus GI/II	Not Detected		
Rotavirus A	Not Detected		
Sapovirus (I,II,IV,V)	Not Detected		
Astrovirus (hAstro)	Not Detected		

Normal Bacterial GUT Flora	Result	Range	Units
Bacteroides fragilis	77.0	1.6 - 250.0	x10^9 CFU/g
Bifidobacterium species	4001.0	> 6.7	x10^7 CFU/g
Bifidobacterium longum	8.8	> 5.2	x10^6 CFU/g
Enterococcus species	36.1	1.9 - 2000.0	x10^5 CFU/g
Escherichia species	3699.0	3.7 - 3800.0	x10^6 CFU/g
Lactobacillus species	767.7	8.6 - 6200.0	x10^5 CFU/g
Lactobacillus Rhamnosus	12.0	8.3 - 885.0	x10^4 CFU/g
Clostridium species	25.7	5.0 - 50.0	x10^6 CFU/g
Enterobacter species	4.0	1.0 - 50.0	x10^6 CFU/g
Akkermansia muciniphila	48.80	1.00 - 50.00	x10^3 CFU/g
Faecalibacterium prausnitzii	386.7	200.0 - 3500.0	x10^3 CFU/g

Methodology:
GIT Functional markers performed by GCMS, EIA,FEIA.
Bacteriology, Virology, Fungi, Parasites & Worms performed by PCR,qPCR.
<dl = result below detectable limit. *H = Result greater than the reference range. *L = Result less than the reference range

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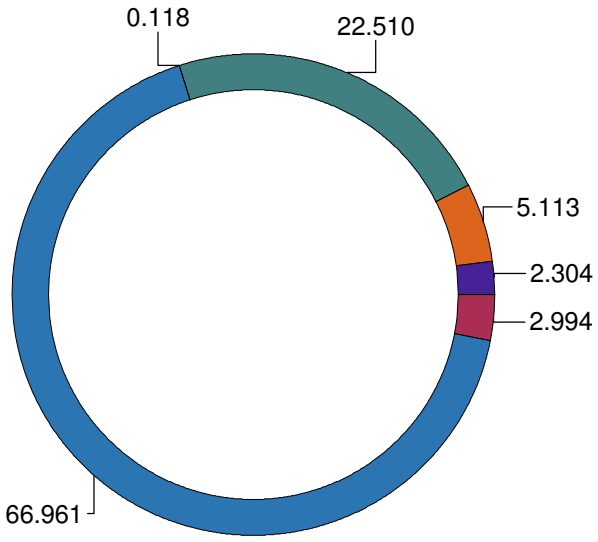
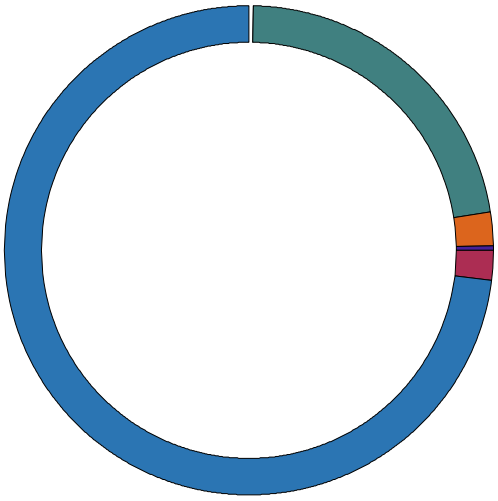
Introduction:
Your gut microbiome is a collective name for the 40 trillion cells and up to 1000 microbial species that include bacteria, viruses, fungi, parasites, and archaea and reside in our gut. The number of gut bacterial cells is approximately equal to the total number of human cells in our body, so if we consider only cell counts, we are only about half human. In terms of gene counts, the microbiome contains about 200 times more genes than the human genome, making bacterial genes responsible for over 99% of our body's gene content! Of all the microbial communities in the human body, the gut microbiome is by far the most dense, diverse, and physiologically important ecosystem to our overall health.

Relative Commensal Abundance	Result	Range	Units
Bacteroidetes Phylum	66.961	50.000 - 95.000	%
Firmicutes Phylum	22.510	3.500 - 40.000	%
Proteobacteria Phylum	5.113	0.500 - 12.500	%
Actinobacteria Phylum	2.994	0.001 - 4.818	%
Verrucomicrobia Phylum	2.304	0.000 - 2.400	%
Euryarchaeota Phylum	0.118	0.000 - 0.177	%

Your Phyla:



Healthy Phyla:



References:
NOTE: Relative abundance reference ranges have been based on a healthy population study.
King CH, et., al. (2019) Baseline human gut microbiota profile in healthy people and standard reporting template. PLoS One. 2019 Sep 11;14(9):e0206484.

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Pathogen Summary:

Macroscopy Comment

BROWN coloured stool is considered normal in appearance.

Faecal Occult Blood Negative:
Faecal occult blood has not been detected in this specimen. If the test result is negative and clinical symptoms persist, additional follow-up testing using other clinical methods is recommended.

Metabolism Comment

In a healthy gut Short Chain Fatty Acids (SCFAs) exhibited in the following proportions; Butyrate, Acetate, Propionate (16%:60%:24%).
The primary SCFAs butyrate, propionate and acetate are produced by predominant commensal bacteria via fermentation of soluble dietary fibre and intestinal mucus glycans.

Key producers of SCFAs include Faecalibacterium prausnitzii, Akkermansia mucinphila, Bacteroides fragilis, Bifidobacterium, Clostridium and Lactobacillus Spp.
The SCFAs provide energy for intestinal cells and regulate the actions of specialised mucosal cells that produce anti-inflammatory and antimicrobial factors, mucins that constitute the mucus barriers, and gut active peptides that facilitate appetite regulation and euglycemia. Abnormal SCFAs may be associated with dysbiosis, intestinal barrier dysfunction and inflammatory conditions.

SCFA PRODUCTION TABLE

BACTERIA	BUTYRATE	PROPRIONATE	ACETATE
Akkermansia muciniphila		▲▲▲	▲▲
Anaerostipes caccae	▲▲▲		
Bacteroides spp.		▲▲▲	
Bifidobacterium spp.		▲	▲▲▲
Blautia obeum		▲▲	▲▲▲
Coprococcus eutactus	▲		
Escherichia coli			▲
Eubacterium rectale	▲▲		
Faecalibacterium prausnitzii	▲▲▲		
Lactobacillus spp.	▲	▲	▲
Roseburia homini	▲▲		
Ruminococcus bromii	▲		▲
Subdoligranulum variabile	▲		

KEY	
▲	Low Producers
▲▲	Moderate Producers
▲▲▲	High Producers

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GIT Markers Comment

PANCREATIC ELASTASE: MILD TO MODERATE INSUFFICIENCY.
Pancreatic insufficiency reflects trypsin, chymotrypsin, amylase and lipase activity.
PE1 is also useful in monitoring exocrine pancreatic function caused by: Chronic pancreatitis, Autoimmunopathies & connective tissue diseases, Chronic inflammatory bowel disease, Intestinal malabsorption with mucosal atrophy.
Treatment:
- Digestive enzyme supplementation
- A low-fat diet to control steatorrhea (excess fat in stools)
- Vitamin and mineral supplementation
- Investigate underlying causes for reduced pancreatic function (for eg. Coeliac disease, duodenal enteropathy, pancreatitis).

CALPROTECTIN Normal:
Faecal calprotectin values <50 ug/g are not indicative of inflammation in the gastrointestinal tract. Subjects with low faecal calprotectin levels normally do not need to be further investigated by invasive procedures. In patients with strong clinical indications of intestinal inflammation, repeat testing may be useful.
Test performed by Phadia EliA Fluorescence enzyme immunoassay (FEIA).

LOW SECRETORY IGA:
Secretory IgA represents the first line of defence of the gastrointestinal mucosa and is central to the normal function of the gastrointestinal tract as an immune barrier.

Secretory IgA binds to invading microorganisms and toxins and entrap them in the mucus layer or within the epithelial cells, so inhibiting microbial motility, agglutinating the organisms, and neutralising their exotoxins and then assist in their harmless elimination from the body in the faecal flow. sIgA also 'tags' food as acceptable, so low sIgA leads to increased sensitivity to foods. Several studies link stress and emotionality with levels of sIgA. Production is adversely affected by stress, which is mediated by cortisol levels.
Often low levels of Secretory IgA correlates with low beneficial flora levels and an increase in pathogenic and parasitic organism being present.

Treatment: Investigate the root cause and rule out parasitic organisms or pathogenic bacteria. Consider the use of probiotics (saccharomyces boulardii), choline, essential fatty acids, glutathione, glycine, glutamine, phosphatidylcholine, Vitamin C and Zinc which are all required for efficient production of Secretory IgA.
PLEASE NOTE: A low Secretory IgA should be reviewed in conjunction with the stool formation. An artefactually low level may be due to fluid dilution effects in a watery or unformed/loose stool sample.

ELEVATED ZONULIN LEVELS:
Zonulin is a protein that modulates intestinal barrier function and can also be considered as a potential inflammatory marker. Review other markers in conjunction to this result such as faecal calprotectin.
Zonulin release facilitates the opening of tight junctions between the cells of the intestinal lining to allow for passage of nutrients and fluids into the body. However, Zonulin release can be "overstimulated" by certain external factors to cause excessive opening of tight junctions, leading to intestinal hyperpermeability or "leaky gut", inflammation, liver overload, nutrient deficiencies, rheumatoid arthritis and autoimmune disorders.
Identify the possible cause/s (Gut microorganism imbalance or the presence of dietary Gluten/gliadin) and remove to reduce further damage.

beta GLUCURONIDASE ELEVATED:
Beta-glucuronidase is a bacterial enzyme that may limit the body's ability to excrete compounds such as drugs, hormones, and environmental toxins. Certain bacteria may also increase Beta-glucuronidase such as elevated levels of E.coli.

Treatment:
Consider Calcium-D-glucarate which may assist with lowering B-glucuronidase levels. It is also suggested to introduce a low-calorie/vegetarian diet for 4 weeks which may also be beneficial with lowering faecal B-glucuronidase levels. Additionally, one human study has suggested that consuming glucomannan can reduce fecal beta-glucuronidase activity. Glucomannan is a type of prebiotic fiber found in konjac root which is commonly used to make low calorie pasta and noodles.

Opportunistic Bacteria Comment

METHANOBACTERIACEAE ELEVATED:

PHYLUM: Euryarchaeota

DESCRIPTION:
Methanobacteriaceae are methane-producing microbes that play an important role in the gut ecosystem by facilitating carbohydrate

fermentation and production of short-chain fatty acids by commensal bacteria. Elevated levels may be associated with abdominal bloating, constipation, flatulence, inflammatory bowel disease (IBD), irritable bowel syndrome (IBS), colorectal cancer, diverticulosis or obesity and often correlate with a positive SIBO test.

TREATMENT SUGGESTIONS:
Elimination of methanogenic flora using antibiotic treatment may contribute to therapeutic benefits and include neomycin or rifaximin. Antimicrobial herbs may also be beneficial in treatment (including garlic and oregano). Rule out allergy to above medication before prescribing/taking.

Bacterial Pathogens Comment

ENTEROTOXIGENIC ESCHERICHIA COLI LT/ST ELEVATED:

PHYLUM: Proteobacteria

DESCRIPTION:
Enterotoxigenic E. coli (ETEC) are a pathogenic variant or pathovar of E. coli defined by production of diarrheagenic heat-labile (LT) and heat-stable (ST) enterotoxins. It is transmitted by ingestion of contaminated food or water. Enterotoxigenic E. Coli is the most common cause of traveller's diarrhoea (watery). It is estimated that up to 3 % of individuals may have symptoms that last more than one month, and that post-infectious IBS occurs in an estimated 10-30 % of patients.

TREATMENT SUGGESTIONS:
The treatment of diarrheal disease due to ETEC is the same as that for cholera or any other acute secretory diarrheal disease. The correction and maintenance of hydration is always most important. Although ETEC infections are typically self-limited, antibiotics can hasten the resolution of symptoms. Antibiotics that have been used in effective treatment include doxycycline, trimethoprim-sulfamethoxazole, erythromycin, norfloxacin, ciprofloxacin, ofloxacin, azithromycin, and rifamycin. Rule out allergy to above medication before prescribing/taking. Consult ID specialist if patient is showing severe symptoms or immunocompromised.

Normal Bacterial Flora Comment

LACTOBACILLUS SPECIES LOW NORMAL:

PHYLUM: Firmicutes

DESCRIPTION:
Lactobacillus species is below average.

Lactobacillus is a genus of Gram-positive rod-shaped, non-spore-forming bacteria which constitute a significantly important component of the human gastrointestinal system.

Lactobacillus exhibits a mutualistic relationship with the human body, as it protects the host against potential invasions by pathogens, and in turn, the host provides a source of nutrients. Lactobacilli are among the most common probiotic found in food such as yogurt, and it is diverse in its application to maintain gut health and to help treat ailments clinically such as diarrhea and obesity.

Lactobacillus species promote the anti-inflammatory response, thereby supporting the improvement of symptoms pertaining to asthma, chronic obstructive pulmonary disease, neuroinflammatory diseases, cardiovascular diseases, inflammatory bowel disease (IBD) and chronic infections in patients. Reduced levels increase the risk of infections and inflammation. Studies have also revealed that chronic psychological stress and alcohol use may be associated with a decrease in Lactobacillus species, as well as antibiotic / medication use.

TREATMENT SUGGESTIONS: Treatment may involve the use of Lactobacillus containing probiotics and treatment of any intestinal infections.

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The Four "R" Treatment Protocol

REMOVE	Using a course of antimicrobial, antibacterial, antiviral or anti parasitic therapies in cases where organisms are present. It may also be necessary to remove offending foods, gluten, or medication that may be acting as antagonists. Consider testing IgG96 foods as a tool for removing offending foods.	ANTIMICROBIAL	Oil of oregano, berberine, caprylic acid
		ANTIBACTERIAL	Liquorice, zinc carnosine, mastic gum, tribulus, berberine, black walnut, caprylic acid, oil of oregano
		ANTIFUNGAL	Oil of oregano, caprylic acid, berberine, black walnut
		ANTIPARASITIC	Artemesia, black walnut, berberine, oil of oregano
		ANTIVIRAL	Cat's claw, berberine, echinacea, vitamin C, vitamin D3, zinc, reishi mushrooms
		BIOFILM	Oil of oregano, protease
REPLACE	In cases of maldigestion or malabsorption, it may be necessary to restore proper digestion by supplementing with digestive enzymes.	DIGESTIVE SUPPORT	Betaine hydrochloride, tilactase, amylase, lipase, protease, apple cider vinegar, herbal bitters
REINOCULATE	Recolonisation with healthy, beneficial bacteria. Supplementation with probiotics, along with the use of prebiotics helps re-establish the proper microbial balance.	PREBIOTICS	Slippery elm, pectin, larch arabinogalactans
		PROBIOTICS	Bifidobacterium animalis sup lactise, lactobacillus acidophilus, lactobacillus plantarum, lactobacillus casei, bifidobacterium breve, bifidobacterium bifidum, bifidobacterium longum, lactobacillus salivarius ssp salivarius, lactobacillus paracasei, lactobacillus rhamnosus, Saccaromyces boulardii
REPAIR & REBALANCE	Restore the integrity of the gut mucosa by giving support to healthy mucosal cells, as well as immune support. Address whole body health and lifestyle factors so as to prevent future GI dysfunction.	INTESTINAL MUCOSA IMMUNE SUPPORT	Saccaromyces boulardii, lauric acid
		INTESTINAL BARRIER REPAIR	L-Glutamine, aloe vera, liquorice, marshmallow root, okra, quercetin, slippery elm, zinc carnosine, Saccaromyces boulardii, omega 3 essential fatty acids, B vitamins
		SUPPORT CONSIDERATION	Sleep, diet, exercise, and stress management